
CharXiv: Charting Gaps in Realistic Chart Understanding in Multimodal LLMs

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Abstract

Chart understanding plays a pivotal role when applying Multimodal Large Language Models (MLLMs) to real-world tasks such as analyzing scientific papers or financial reports. However, existing datasets often focus on oversimplified and homogeneous charts with template-based questions, leading to an overly optimistic measure of progress. We demonstrate that although open-source models can appear to outperform strong proprietary models on these benchmarks, a simple stress test with slightly different charts or questions can deteriorate performance by up to 34.5%. In this work, we propose CharXiv, a comprehensive evaluation suite involving 2,323 natural, challenging, and diverse charts from arXiv papers. CharXiv includes two types of questions: 1) *descriptive* questions about examining basic chart elements and 2) *reasoning* questions that require synthesizing information across complex visual elements in the chart. To ensure quality, all charts and questions are handpicked, curated, and verified by human experts. Our results reveal a substantial, previously underestimated gap between the reasoning skills of the strongest proprietary model (i.e., GPT-4o), which achieves 47.1% accuracy, and the strongest open-source model (i.e., InternVL Chat V1.5), which achieves 29.2%. All models lag far behind human performance of 80.5%, underscoring weaknesses in the chart understanding capabilities of existing MLLMs. We hope that CharXiv facilitates future research on MLLM chart understanding by providing a more realistic and faithful measure of progress.

1 Introduction

Multimodal Large Language Models (MLLMs) [4, 56, 14, 82, 54, 16, 17, 12, 40, 41, 7, 2, 5, 73, 78, 49] are highly versatile and effective for a wide range of real-world applications [64, 69, 20, 57, 92, 61, 68, 60, 93]. Within these applications, chart understanding is a highly desired capability as charts are ubiquitous in scientific papers, financial reports, and news articles. It also poses unique challenges where models need to perform complex reasoning over numerical data, textual labels, and complex visual elements to answer difficult questions (see Fig. 1), thus making chart understanding a suitable measure of progress for MLLMs. Many benchmarks in the popular MathVista evaluation suite [60] are designed to test chart understanding. However, these benchmarks lack diversity in both the types and complexity of the charts and the often template-based questions (§2.1). For example, FigureQA [36] and DVQA [35] rely on procedurally generated question templates. While ChartQA [64] includes a mixture of handwritten and machine-generated questions, the charts lack visual diversity due to the homogeneous appearance of the charts from a limited number of sources. Regardless, many proprietary models [2, 78, 5, 73] and open-source models [12, 17, 16, 41, 31, 49, 55, 21] are evaluated

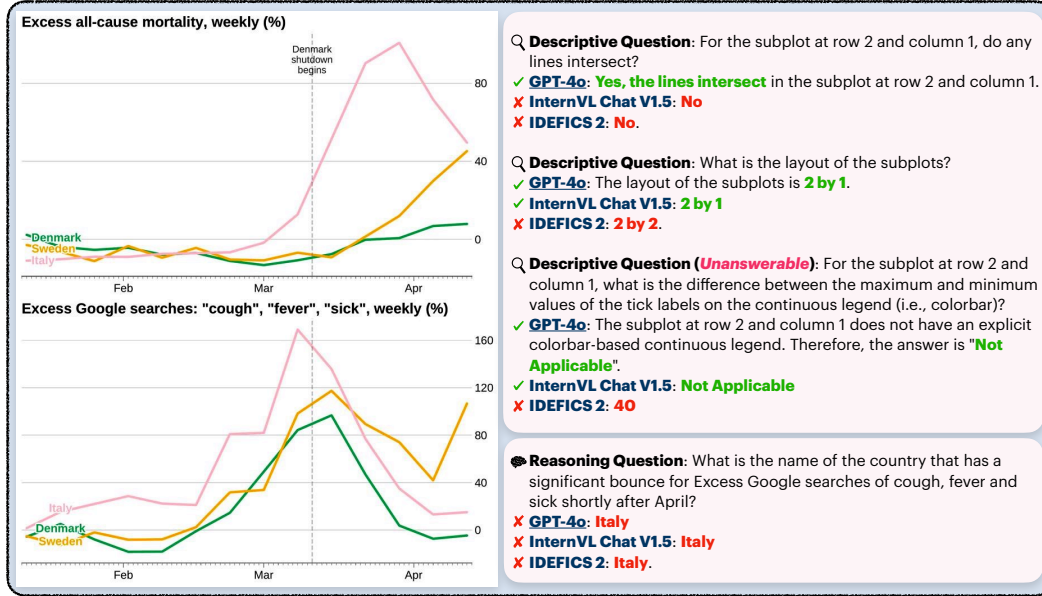


Figure 1: Example chart (left), descriptive questions (top-right) and reasoning questions (bottom-right) in CharXiv where open-source models even fail in basic descriptive questions. Moreover, all models struggle with correctly answering the reasoning question.

on these datasets.¹ These narrow evaluations create the appearance that the open-source models outperform proprietary ones², despite evidence to the contrary: we designed simple stress tests (§2.2) in which we find that open-source models lag far behind proprietary ones in their robustness to small visual or textual changes. For example, the accuracy of SPHINX V2 dropped from 63.2% to 28.6% with a 34.5% gap when questions are slightly modified with respect to the same set of charts.

We introduce CharXiv, a comprehensive evaluation suite for complex understanding of natural, challenging, and diverse charts (§3) to address the above issue. CharXiv consists of 2,323 real-world charts handpicked from scientific papers spanning 8 major subjects published on arXiv (§3.1). We explicitly disentangle visual understanding and reasoning by designing two types of questions (§3.2): (1) *descriptive* questions, requiring understanding basic chart information such as the title, labels, and ticks; (2) *reasoning* questions, requiring comparisons, approximations, and fine-grained analysis. CharXiv is an especially high-quality dataset where all questions are *manually* curated by human experts, and all ground-truth answers are validated by hand. To answer both types of questions, the model only needs to understand the visual contents of the chart without advanced domain-specific knowledge and contextual information. Evaluating an MLLM on CharXiv is straightforward, because we impose a short answer format that is amenable to LLM-based automatic grading.

We extensively evaluate 13 open-source models and 11 proprietary models (§4.1) and identify a large disparity between the strongest open-source and proprietary models (§4.2): InternVL Chat V1.5 correctly answers only 29.2% of the reasoning questions and 58.5% of the descriptive ones, whereas GPT-4o correctly answers 47.1% of the reasoning questions and 84.5% of the descriptive ones (Tab. 3). As shown in Fig. 2, the performance gap in the reasoning questions of 17.9% is significantly larger than the gap identified in prior works [35, 36, 64]. Further, both types of models lag far behind the human performance of 80.5% on the reasoning questions and 92.1% on the descriptive ones. Fine-grained analysis of model performance (§4.3) shows several insights owing to the design of CharXiv. In particular, we characterize: (1) differences in reasoning and descriptive capabilities, exploring when one skill reinforces the other; (2) what types of tasks and charts are difficult for existing MLLMs; (3) how different MLLMs respond to unanswerable questions. Overall, we hope that CharXiv enables a thorough, multi-faceted evaluation of chart understanding in MLLMs.

¹We note that there are several more sophisticated benchmarks [86, 85, 53] that have recently been released. We discuss key differences between CharXiv and these benchmarks in §2.1.

²See the FQA (i.e., Figure QA) column of the MathVista [leaderboard](#). Throughout the paper, “open-source” refers to models with publicly available weights.

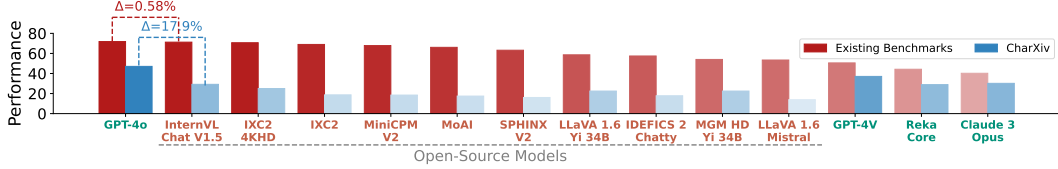


Figure 2: Model performance comparison on reasoning questions from [CharXiv](#) v.s. questions from existing benchmarks. As indicated by the red and blue bars respectively, many open-source models surpass proprietary model performance on the 174 sample questions from existing benchmarks (subsets of DVQA, FigureQA and ChartQA from the *testmini* split of MathVista) yet fail consistently on the 1000 reasoning questions from the validation split of [CharXiv](#).

2 Existing Benchmarks Overestimate Chart Understanding Capabilities

2.1 Related Works

Existing benchmarks such as FigureQA [36], DVQA [35], PlotQA [71] do not fully capture the complexity and diversity of real-world charts due to their synthetic nature, while charts in ChartQA [64] lack visual diversity. More recent benchmarks such as MMC [53], ChartBench [86] and ChartX³ [85] also contain issues with the source or diversity of the charts (e.g., ChartX, MMC) and the types of questions (e.g., MMC, ChartBench). We provide a summary of existing benchmarks’ design choices in Tab. 1 and a detailed review below. We provide a more detailed related works on Multimodal Large Language Models and More MLLM benchmarks in App. B.

Chart source. FigureQA, DVQA and PlotQA use plotting software to synthesize charts restricted to very few predefined chart types with stylistically similar elements (see Figs. 8(a), 8(b) and 8(c)). ChartQA sources charts from only 4 websites, each of which lacks visual diversity (see Fig. 8(d)). One such website also served as the primary source of charts for reasoning questions in MMC. On the other hand, ChartX provides fixed instructions to GPT-4 to write code to procedurally generate predefined types of charts and settings in bulk. All of these approaches yield artificial charts belonging to a narrow distribution.

Question types. Existing benchmarks lack variation in their questions: FigureQA, DVQA and PlotQA use a fixed template to generate QA pairs, while

Table 1: Design choice of chart understanding benchmarks. We use the following shorthand: Vis. Div.=visual diversity, Temp.=template, Knwl.=knowledge, and Vocab.=vocabulary. Cells marked with “✓” indicate *mixed attributes* (e.g., real and synthetic data; real and synthetic chart).

Name	QUESTION TYPE						ANSWER
	Real Data	Real Chart	Vis. Div.	Temp. Based	Free Form	Knwl. Free	Open Vocab.
<i>QA-Based</i>							
FigureQA [36]	✗	✗	✗	✓	✗	✓	✗
DVQA [35]	✗	✗	✗	✓	✗	✓	✓
PlotQA [71]	✓	✗	✗	✓	✗	✓	✓
ChartQA [64]	✓	✓	✗	✗	✓	✓	✓
ChartBench [86]	✓	✓	✓	✓	✗	✓	✗
<i>Multi-Task</i>							
MMC [53]	✓	✓	✗	✗	✓	✗	✓
ChartX [85]	✗	✗	✓	✗	✓	✓	✓
CharXiv	✓	✓	✓	✓	✓	✓	✓

ChartBench adopts an automatic QA generation pipeline according to 4 predefined tasks. However, similar to MMMU [93], more complex reasoning questions from MMC cannot be solved from the charts alone and require external domain-specific knowledge (e.g., mapping acronyms in the legend to particular algorithms).

Answer & validation. FigureQA and ChartBench both evaluate model performance based only on *yes/no* questions. Evaluating models on binary answers does not faithfully reflect their performance in the natural use case of general free-form question answering [48].

2.2 Open-Source MLLMs Are Sensitive to Perturbations

Many open-source models have adapted the training sets of existing benchmarks [36, 35, 64] for visual instruction tuning [56] and show promising performance in their respective evaluation sets. However, due to the aforementioned issues with the diversity of these benchmarks, the evaluation

³Due to limited public availability of the MMC and ChartBench data, our assessment is based on the papers.

data is too similar to the training data. As a result, evaluation scores often do not accurately reflect the general chart understanding capabilities of MLLMs. In particular, we demonstrate below that *simple* modifications in the evaluation components lead to *drastic* changes in model performance.

Models. We selected open-source models that are known to be trained on the training set of DVQA and ChartQA: Mini-Gemini (MGM) [49], InternVL-XComposer2 (IXC2) [16], InternVL-XComposer2 4KHD (IXC2 4KHD) [17], InternVL-Chat V1.5 [12], SPHINX V2 [21], LLaVA 1.6 [55], and IDEFICS 2 [41]. We compare their performance with proprietary models [2, 5, 73].

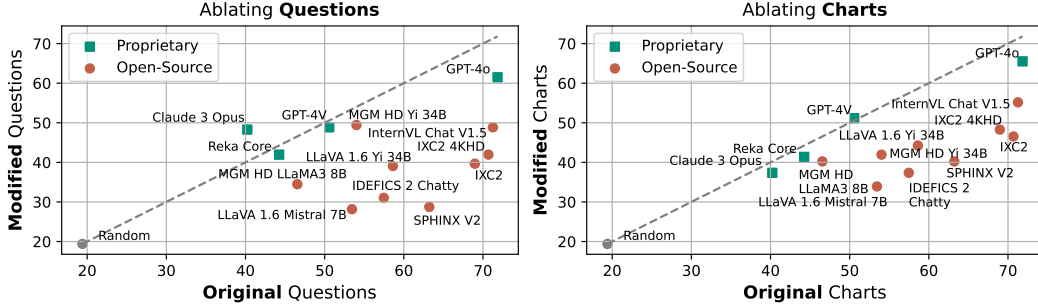


Figure 3: Open-source models generalize poorly to modified examples (measured by accuracy). Left: original set against modified-*question* set. Right: original set against modified-*chart* set.

Evaluation set. We extracted subsets of DVQA, FigureQA, and ChartQA from MathVista. This yields 174 samples, and we refer to it as the *original set*. To test the robustness of the models mentioned above, we created two modified versions of the original set: the modified-question set (see App. S) and the modified-chart set (see App. T). In the modified-question set, we retain the original chart, but write novel questions that deviate from the predefined templates [36, 35]. In the modified-chart set, we alter the charts to those from arXiv with similar visual complexity that can be asked with the same types of questions. We manually annotate all questions and answers in both the modified-question and the modified-chart set. As in the original set, we maintain an equal number of yes and no answers in the original set to prevent models from achieving artificially high scores by simply outputting one response more often than the other, and adopt the same evaluation protocol as in MathVista.

Results. As plotted in Fig. 3, all proprietary models remain close to the diagonal line, indicating good generalization in both modified-question and modified-chart scenarios. In contrast, most open-source models exhibit significant performance degradation in both settings, indicating poor generalization. We observe the most pronounced performance drop in SPHINX V2 in the modified-question set, where performance dropped by 34.5%, from 63.2% in the original set to 28.7% in the modified-question set. Our findings demonstrate that design strategies in existing benchmarks lead to an *overestimation* of chart understanding capabilities for open-source models. We hypothesize that the training and evaluation datasets are too similar, so models appear to generalize well despite not being robust to simple modifications. In the next section, we introduce **CharXiv**, which features a more natural, challenging, and diverse evaluation of real-world charts.

3 CharXiv: A Real-World and Challenging Chart Understanding Benchmark

CharXiv is a comprehensive and challenging chart understanding benchmark sourced solely from real-world charts. We select diverse, naturally occurring, and complex figures from arXiv preprints, and manually construct descriptive and reasoning questions that require intensive visual and numerical analysis. **CharXiv** consists of 2,323 charts paired with more than 10K questions—we randomly sample 1,000 charts as the validation set and use the rest as the test set.⁴ In the following sections, we describe how we select charts (§3.1), construct questions (§3.2), and validate model responses (§3.3).

⁴Similar to MathVista [60] and MMMU [93], we release all QA pairs for the validation set and keep the answers to the test set private to prevent data leakage.

Chart Metadata		Reasoning Questions		Descriptive Questions	
Categories		Answer Type		Information Extraction	
Computer Science (292; 12.6%)		Text in Chart (1044; 45%)		Title (591; 6.4%)	
Economics (287; 12.3%)		Number in Chart (512; 22.0%)		x-axis Label (519; 5.6%)	
Elec. Eng. & Sys. Sci. (291; 12.5%)		Text in General (229; 9.9%)		y-axis Label (494; 5.3%)	
Mathematics (286; 12.3%)		Number in General (538; 23.2%)		Leftmost Tick (586; 6.3%)	
Physics (294; 12.7%)		QA Source		Rightmost Tick (581; 6.3%)	
Quant. Biology (293; 12.6%)		GPT-Generated (448; 19.3%)		Lowest Tick (570; 6.1%)	
Quant. Finance (289; 12.4%)		GPT-Inspired (497; 21.4%)		Highest Tick (537; 5.8%)	
Statistics (291; 12.5%)		Human-Written (1378; 59.3%)			
				Pattern Recognition	
Year				Line Intersection (358; 3.9%)	
2020 (581; 25.0%)				Trend of Data (85; 0.9%)	
2021 (585; 25.2%)				Subplot Layout (566; 6.1%)	
2022 (584; 25.1%)					
2023 (573; 24.7%)					
Number of Subplots		Answerability		Counting	
Single (896; 28.6%)		Answerable (6969; 75%)		# Lines (324; 3.5%)	
2-4 (876; 37.7%)		Unanswerable (2323; 25%)		# Labels (471; 5.1%)	
5+ (551; 23.7%)				# Subplots (144; 1.6%)	
				Compositionality # tick labels across all axes (566; 6.1%)	

Figure 4: Metadata breakdown of charts, descriptive questions, and reasoning questions in CharXiv.

3.1 Chart Curation

Figure source. We downloaded all arXiv preprints on eight academic subjects from January 2020 to September 2023 (Fig. 4) and extracted figures from the source files. All figures were re-rendered into high-resolution JPEG format, with the longer side of each figure resized to 1024px.

Chart selection. We define a chart as *any figure that visually illustrates data*. Most figures in arXiv source files are diagrams, illustrations, and natural images, *not* charts. To identify charts and promote visual diversity, we apply a four-step selection pipeline. First, we utilize a pretrained SigLIP visual encoder [94] to identify candidate figures that exhibit a cosine similarity of at least 0.65 with the average image embedding of existing charts from MathVista [35, 36, 64, 60]. We choose this target similarity to balance identifying charts and ensuring good coverage of the visually diverse distribution. Second, we recruit experienced graduate students to manually select charts from the candidate set. Concretely, we randomly sample 750 candidate figures from the pre-filtered set for each subject and year, and present 10 figures at a time to the annotators, asking them to select a single figure that is a chart and looks different from previously selected datapoints (see App. R.1 for details). In the third step, we remove the charts that exhibit large (≥ 0.95) pairwise cosine similarities with the other candidates. Finally, we remove the charts that are not clearly labeled or appear blurry. At the end of this four-step pipeline, we have 2,323 charts in total.

We provide details of the chart categories, years, and number of subplots in Fig. 4, size information in Tab. 2, and a collage of sampled charts in Fig. 8(e). Notably, the charts in CharXiv exhibit far greater compositional and stylistic complexity compared to those in existing datasets. A single chart often combines elements or subplots featuring multiple chart types (e.g., lines and bars in one plot). To aggregate statistics on chart types, we first query GPT-4o to generate potential chart types for each chart. Human annotators then review and refine this list, assigning a primary chart type based on the chart’s most salient features. We provide chart type statistics in Fig. 5.

3.2 Question Construction

We construct two types of questions: *descriptive* and *reasoning*. Descriptive questions assess models’ capability in extracting and aggregating basic information from charts, and reasoning questions evaluate a model’s ability to perform complex visual reasoning.

Descriptive questions. We designed a total of 19 templates for descriptive questions that require (1) identifying basic information, such as the title, axis labels, legend labels, labeled ticks, or (2) aggregating chart information to count ticks, recognize data patterns, and enumerate labels. These questions are broadly categorized into five groups: information extraction, enumeration, pattern recognition, counting, and compositionality (see App. O.1 for details). Although descriptive questions are intended to be easier than reasoning questions, they can still pose challenges due to the complexity of the charts. For example, answering descriptive questions about charts with multiple subplots requires the model to first identify the relevant subplot⁵ (see Apps. U.1, U.7 and U.10). If basic elements such as the legend, axis, and title are shared across multiple subplots, the model must then also grasp the relationships among the subplots to extract the correct information (see Apps. U.3 and U.23). We pair each chart with four descriptive questions and one of them is intentionally

⁵We use the prefix “for the subplot at row N and column M ” when subplots form a grid or a description e.g., “for the bottom left subplot” otherwise. Both N and M start from 1.